
From Inverse Optimization to Sequential Decisions: Conformal Prediction for Inverse Reinforcement Learning

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Abstract

1 Inverse reinforcement learning (IRL) is inverse optimization with a sequential
2 forward problem: demonstrations reveal an unknown reward, which is then op-
3 timized to prescribe a policy. Seen this way, IRL inherits a weakness that Con-
4 formal Inverse Optimization (CIO) identified for one-shot decisions – collapsing
5 a heterogeneous population of demonstrators to one reward point estimate can
6 produce an arbitrarily bad policy. We introduce **CP-IRL**, which transfers CIO’s
7 calibrate-then-robustify principle to Markov decision processes: from a population
8 of demonstrators, CP-IRL conformally calibrates a set of reward parameters con-
9 sistent with a new demonstrator, then optimizes a policy against that ambiguity.
10 Two pieces of new machinery make this tractable. First, the Bellman resolvent
11 reduces each calibration score to a reward-dimensional (rather than reward-plus-
12 value-function-dimensional) convex program. Second, we show the robust MDP is
13 convex whenever ambiguity is confined to the reward and the transition kernel is
14 fixed – for any compact ambiguity set – and that this boundary is tight: a two-state
15 counterexample loses convexity the moment ambiguity also touches the dynamics.
16 Building on this machinery, we prove finite-sample coverage, a regret bound, and
17 a separation result where point-estimate IRL’s regret grows without bound while
18 CP-IRL’s vanishes. Experiments on random MDPs, gridworld, and Objectworld
19 then locate the method’s actual scope: robust hedging cuts true-reward regret under
20 systematic estimation bias, but costs regret when the point estimate is merely noisy
21 and already unbiased.

22 1 Introduction

23 Inverse reinforcement learning (IRL) is usually introduced as its own problem: infer a reward from
24 behavior, then optimize that reward to get a policy. But end to end, IRL is a sequential instance of
25 *inverse optimization* – the observed decision is a policy or trajectory, the latent objective is a reward
26 function, and the forward problem is an MDP. Under that view, uncertainty about the recovered reward
27 should propagate into the downstream decision instead of being discarded once a point estimate is fit.

28 Conformal Inverse Optimization (CIO) builds exactly this pipeline for one-shot decisions Lin et al.
29 [2024]: it replaces a single recovered objective with a conformally calibrated uncertainty set and
30 optimizes robustly over that set, with a finite-sample, distribution-free guarantee under exchangeabil-
31 ity. CIO’s own separation example shows the stakes – point-estimate inverse optimization can carry
32 unbounded downstream regret even with unlimited data. This raises the question we answer here:
33 can CIO’s calibrate-then-robustify principle transfer to inverse reinforcement learning without losing
34 coverage or tractability?

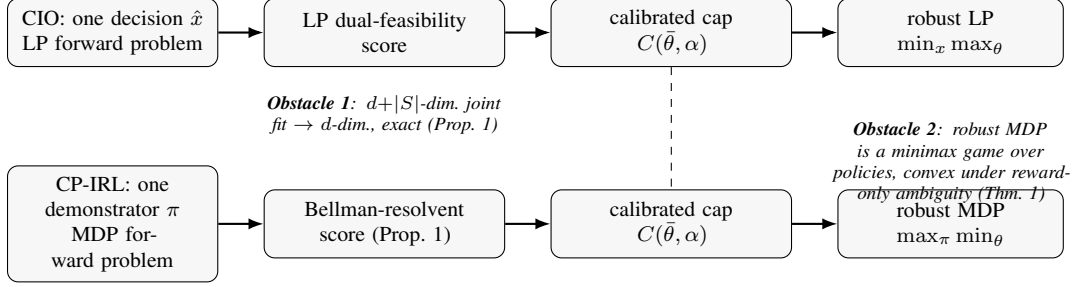


Figure 1: The transfer at a glance. CP-IRL keeps CIO’s calibrate-then-robustify template (same calibrated cap $C(\theta, \alpha)$, dashed link) but replaces every sequential-specific piece: the nonconformity score (§4.1) and the robust decision problem (§4.2).

Our answer, CP-IRL, keeps CIO’s pipeline and rebuilds its machinery for sequential decisions. A population of demonstrators supplies the exchangeable calibration units; each observed policy defines a feasible reward set through Bellman optimality; split conformal prediction calibrates a spherical cap around an IRL point estimator; a robust MDP turns the calibrated ambiguity into a deployed policy. Two obstacles block a direct transplant of this template from LPs to MDPs. First, CIO’s per-point calibration score is computable in closed form because the forward problem is a linear program, so reward feasibility reduces to LP dual feasibility. An MDP has no such shortcut: “this policy is optimal for this reward” is a Bellman-optimality condition, and checking it naively means jointly optimizing over the reward *and* the policy’s value function, a variable count that grows with the state space. Second, CIO’s robust forward problem is a single robust linear program, but the MDP analogue is a minimax game over policies, $\max_{\pi} \min_{\theta}$, over an ambiguity set – and robust MDPs with ambiguity that couples across states are NP-hard in general Wiesemann et al. [2013], with tractable results typically relying on *rectangularity* assumptions that restrict how the adversary can vary its choice by state.

We resolve both obstacles. Figure 1 lays out the transfer: same calibrate-then-robustify template on both rows, with the two obstacles resolved where the sequential setting departs from CIO’s one-shot LP. The contributions are:

- **Transfer: conformal inverse optimization for IRL.** We formulate one demonstrator as one exchangeable inverse-optimization instance, calibrate a reward uncertainty set through Bellman-optimal reward feasibility, and establish finite-sample coverage together with a true-reward regret bound (§3–§5).
- **Optimization: sequential structure without state-sized calibration** (§4.1). For a fixed demonstrator policy, the Bellman resolvent $(I - \beta P_{\pi})^{-1}$ makes the induced value function *linear* in the reward parameter – the same identity underlying Ng & Russell’s Ng and Russell [2000] feasible-reward characterization, which we repurpose as a calibration-score reduction. Substituting it shrinks the per-demonstrator calibration score from a joint optimization over (θ, V) (size $d + |S|$) to one over θ alone (size d), exactly, by the policy improvement theorem. We also give a Sherman–Morrison–Woodbury update that recomputes the resolvent in $O(k^2|S| + k^3)$ rather than $O(|S|^3)$ when two demonstrators’ policies differ at only k states.
- **Tractability: a boundary between reward and dynamics ambiguity** (§4.2). The robust MDP is a convex program in occupancy-measure space for any compact reward ambiguity set, with the transition kernel fixed – a Bellman-recursion-free argument that extends the closest sufficiency-side result (Gadot et al. Gadot et al. [2024], who solve the convex L_p -ball case via Hölder duality and policy gradient) to arbitrary compact sets. We show this boundary is tight: a two-state, two-action MDP with a one-dimensional ambiguity set coupling reward and transition probability breaks concavity, with a hand-verifiable rational-valued Jensen violation.
- **Scope: robustness helps under bias, not noise alone.** Our sequential analogue of CIO’s separation example proves classical point-estimate IRL’s regret can diverge while the calibrated policy’s regret vanishes. Across random MDPs, a region-based gridworld, and Objectworld, calibrated robust hedging reduces true-reward regret when the point estimate is

76 systematically biased (as our degenerate construction manufactures), and increases it when
 77 the estimate is merely noisy but unbiased – a precise, falsifiable condition for when the
 78 method pays off, which we report as a core finding rather than bury as a limitation.

79 2 Related Work

80 **Inverse optimization and its calibration.** Inverse optimization recovers parameters of an optimiza-
 81 tion problem from observed decisions (Ahuja & Orlin Ahuja and Orlin [2001]; Chan et al. Chan
 82 et al. [2018]). Classical formulations return a point estimate; more recent work constructs data-driven
 83 uncertainty sets around it (Chenreddy, Bandi & Delage Chenreddy et al. [2022]; Sun, Liu & Li Sun
 84 et al. [2023]), but – as Lin, Delage & Chan Lin et al. [2024] show – none of these give the specific
 85 guarantee we need: that the *next* observed decision’s true parameter is covered with a stated con-
 86 fidence, distribution-free. Conformal Inverse Optimization is the first to close this gap using split
 87 conformal prediction Vovk et al. [2005], and is the paper this work directly extends from one-shot
 88 decisions to MDPs.

89 **Feasible reward sets in IRL.** A separate line of work characterizes the *set* of rewards consistent
 90 with observed IRL data (Poiani et al. Poiani et al. [2024]; Lazzati & Metelli Lazzati and Metelli
 91 [2025]; Skalse et al. Skalse et al. [2022]) and proves PAC-style bounds on recovering it, using it
 92 directly as a robust-MDP ambiguity set. This line shares our target object – a feasible reward set
 93 characterized by Bellman-optimality constraints – but its guarantees are PAC bounds requiring a
 94 generative model of demonstrator sub-optimality; ours require only exchangeability of demonstrators,
 95 no assumption on the *process* by which any individual demonstrator deviates from optimality. The
 96 transfer question this line raises – whether a feasible reward set identified on one MDP transfers to
 97 another (Schlaginhaufen & Kamgarpour Schlaginhaufen and Kamgarpour [2024]) – is a genuinely
 98 open problem for our setting that we do not attempt to close here (see §7).

99 **Robust MDPs and the rectangularity boundary.** Robust MDPs with ambiguity coupling across
 100 states are NP-hard in general (Wiesemann, Kuhn & Rustem Wiesemann et al. [2013]); tractability is
 101 classically recovered via *s*- or *sa*-rectangularity (Iyengar Iyengar [2005]; Nilim & El Ghaoui Nilim
 102 and El Ghaoui [2005]), which restricts the adversary to choose independently per state. Recent work
 103 extends tractable, *non*-rectangular ambiguity specifically for reward-only uncertainty with a fixed
 104 transition kernel: Gadot et al. Gadot et al. [2024] solve the convex L_p -ball case exactly via Hölder
 105 duality and a policy-gradient method with proven convergence; Xu et al.’s Group-Robust MDPs Xu
 106 et al. [2026] handle *feature-wise* (d-rectangular) ambiguity that couples both reward *and* transition,
 107 at the cost of restricting to that rectangular structure. Our sufficiency result (§4.2) covers the same
 108 fixed-kernel, reward-only regime as Gadot et al. but for an *arbitrary* compact ambiguity set (not only
 109 L_p -balls), via a direct occupancy-measure convex program rather than a policy-gradient method; our
 110 necessity result – an explicit demonstration that convexity is lost the moment ambiguity couples into
 111 the kernel, even minimally – has no counterpart in either paper.

112 **Conformal prediction in RL-adjacent settings.** Conformal prediction has been used for offline-RL
 113 pessimism from Q-value or state-action-coverage uncertainty (UARM, Pan et al. Pan et al. [2026]),
 114 for calibrated intervals in infinite-horizon policy evaluation, and for modeling *other agents’ actions*
 115 as conformal prediction sets in multi-agent RL (CAMMARL, Gupta & Kahou Gupta and Kahou
 116 [2023]) – the last of these is explicitly motivated by an analogy to IRL but never calibrates a reward
 117 set. None combine conformal calibration with a reward vector recovered from a population of IRL
 118 demonstrators and a robust-MDP regret guarantee, which is the gap this paper closes.

119 **Preference learning and RLHF.** Preference-based and RLHF-style learning fits a reward (or policy)
 120 from labeled comparisons or demonstrations, typically from individual expert trajectories (Christiano
 121 et al. Christiano et al. [2017]; Rafailov et al. Rafailov et al. [2023]) or crowdsourced preference
 122 populations. Our per-demonstrator exchangeable-unit design (§3) is structurally suited to recast “one
 123 annotator, one calibration point” for heterogeneous-preference RLHF populations. The resulting cov-
 124 erage guarantee is in the intersection-nonemptiness sense of Theorem 2, not parameter containment:
 125 the calibrated reward-ambiguity set is compatible with a γ -fraction of the annotator population’s
 126 individual reward functions, which are never directly observed. We are not aware of this guarantee
 127 elsewhere in the RLHF calibration literature, though we test it only on a toy response-selection MDP,
 128 not at RLHF’s actual scale (§7).

3 Setting

We now make the CIO-IRL correspondence precise. We adopt the population-of-demonstrators setting used throughout this work: a shared tabular MDP $M = (S, A, P, \beta, \phi)$ with known transition kernel P , discount β , and feature map $\phi : S \times A \rightarrow \mathbb{R}^d$ inducing reward $r_\theta(s, a) = \phi(s, a)^\top \theta$. A population of demonstrators draws perceived rewards $\hat{\theta}_1, \dots, \hat{\theta}_N \sim P_\theta$ i.i.d. and each acts optimally in M under their own $\hat{\theta}_k$, producing an observed policy $\pi_k = \pi^*(\hat{\theta}_k)$ – we assume the full deterministic policy is observed exactly, not estimated from a finite trajectory sample; relaxing this to trajectory-level observation would make $\hat{\Theta}(\pi_k)$ itself an estimated (rather than exact) feasible set and is outside the scope of the guarantees below.

Assumption 1 (Exchangeable calibration unit). *The pairs $(\hat{\theta}_k, \pi_k)$, $k = 1, \dots, N, N+1$, are i.i.d. (hence exchangeable), and M (in particular P) is shared and fixed across all demonstrators and the deployment instance.*

Under Assumption 1, each demonstrator plays the role of one (decision, context) pair in CIO’s setting, and it is the setting under which CIO’s coverage argument transplants essentially verbatim (§5, Theorem 2). We commit to this “unit A” design over two weaker alternatives – per-task exchangeability with a single demonstrator (still exchangeable, treated as an easy corollary) and per-trajectory exchangeability within one demonstrator (broken by within-trajectory Markov dependence, and used only as an ablation) – and are explicit throughout about what breaks under each relaxation.

The learning task, mirroring CIO’s pipeline, has three steps: (i) split the population into training and calibration sets; (ii) fit a point estimate $\bar{\theta}$ on the training set with any IRL method (we use and compare a sub-optimality-loss linear program, MaxEnt IRL, and Bayesian IRL by MCMC – §6); (iii) calibrate a spherical-cap uncertainty set $C(\bar{\theta}, \alpha) = \{\theta : \|\theta\|_2 \leq 1, \bar{\theta}^\top \theta \geq \cos \alpha\}$ on the calibration set and solve the robust MDP $\max_\pi \min_{\theta \in C(\bar{\theta}, \alpha)} \theta^\top \mu(\pi)$ in occupancy-measure space, where $\mu(\pi)$ is π ’s occupancy measure.

4 Method

4.1 Tractable calibration via the Bellman resolvent

CIO’s calibration score for a validation point k is $c_k = \max_\theta \{\theta^\top \bar{\theta} : \theta \in \Theta^{\text{OPT}}(\hat{x}_k, u_k), \|\theta\|_2 \leq 1\}$, where Θ^{OPT} is characterized by LP dual feasibility – tractable because the forward problem is a linear program. In an MDP, the analogous feasible set for a policy π is

$$\hat{\Theta}(\pi) = \{\theta : \|\theta\|_2 \leq 1, Q_\theta^\pi(s, a) \leq Q_\pi^\pi(s, \pi(s)) \forall s, \forall a\},$$

the exact set of reward vectors for which π is Bellman-optimal, by the policy improvement theorem. Computing this naively requires the joint optimization $\max_{\theta, V} \theta^\top \bar{\theta}$ subject to the Bellman-optimality inequalities written in terms of both θ and the a priori unknown value function V – a variable count of $d + |S|$.

The key fact we use, which is the sequential-decision restatement of the feasible-reward characterization of Ng & Russell Ng and Russell [2000]: for a **fixed** policy π , the policy’s own induced value function V_θ^π is *linear* in θ , via the Bellman resolvent.

Proposition 1 (Resolvent-based score reduction). *Fix a policy π , let $P_\pi \in \mathbb{R}^{|S| \times |S|}$ be the induced state-transition matrix ($P_\pi(s, s') = P(s, \pi(s), s')$), and let $\Phi_\pi \in \mathbb{R}^{|S| \times d}$ be the feature matrix restricted to the actions π takes ($\Phi_\pi(s, \cdot) = \phi(s, \pi(s))^\top$). Let $W_\pi := (I - \beta P_\pi)^{-1} \Phi_\pi$ (well-defined since $\beta \in [0, 1)$). Then $V_\theta^\pi = W_\pi \theta$ and $Q_\theta^\pi(s, a) = \Psi_\pi(s, a)^\top \theta$ with $\Psi_\pi(s, a) := \phi(s, a) + \beta P(s, a, \cdot)^\top W_\pi$, both linear in θ . Consequently*

$$\hat{\Theta}(\pi) = \{\theta : \|\theta\|_2 \leq 1, \Psi_\pi(s, a)^\top \theta \leq \Psi_\pi(s, \pi(s))^\top \theta \forall s, a\},$$

exactly (no relaxation), and the calibration score $c_k = \max_{\theta \in \hat{\Theta}(\pi_k)} \theta^\top \bar{\theta}$ is an optimization over θ alone – size d , not $d + |S|$ – with W_{π_k} (and hence Ψ_{π_k}) precomputed once per distinct policy via a single resolvent solve.

The proof substitutes the linear Bellman equation’s solution into Q_θ^π and rewrites the policy-improvement optimality condition using Ψ_π ; full proof in Appendix A.1. We checked Proposition 1

175 against a slow joint- (θ, V) reference implementation and against independent value iteration, match-
 176 ing to solver precision on randomized instances; the speedup over the joint formulation grows with
 177 $|S|$, $2\text{--}5.5\times$ at a few hundred states. The Sherman–Morrison–Woodbury update – recomputing the
 178 resolvent in $O(k^2|S| + k^3)$ rather than $O(|S|^3)$ when a new demonstrator’s policy differs from an
 179 already-resolved one at only k states – has no counterpart in Ng & Russell Ng and Russell [2000]; it
 180 is the piece specific to calibrating against a whole population rather than a single demonstrator.

181 4.2 A tight tractability boundary for the robust MDP

182 The robust decision problem, $\max_{\pi} \min_{\theta \in C} \theta^\top \mu(\pi)$, is a minimax game over policies. We work
 183 in occupancy-measure space: $M(P)$ denotes the (fixed, since P is known) polytope of achiev-
 184 able occupancy measures μ , normalized so $\sum_{s,a} \mu(s,a) = 1$, and $\Phi \in \mathbb{R}^{|S \times A| \times d}$ is the ma-
 185 trix whose rows are $\phi(s,a)^\top$ (matching the convention already used for Φ_π in §4.1), so that
 186 $\theta^\top \Phi^\top \mu = \sum_{s,a} \mu(s,a) r_\theta(s,a)$ is the expected per-step reward of occupancy measure μ under
 187 reward θ , and $\theta^\top \Phi^\top \mu / (1 - \beta)$ is the corresponding expected discounted return. The robust objective
 188 is $\max_{\mu \in M(P)} \min_{\theta \in C} \theta^\top \Phi^\top \mu$ (the $1/(1 - \beta)$ factor is a positive constant and does not affect the
 189 arg max).

190 **Theorem 1** (Reward-only ambiguity is convex; this is tight). *(a) Sufficiency. For any nonempty*
 191 *compact ambiguity set $C \subseteq \mathbb{R}^d$ (e.g., our calibrated spherical cap $C(\bar{\theta}, \alpha) = \{\theta : \|\theta\|_2 \leq 1, \theta^\top \bar{\theta} \geq$*
 192 *$\cos \alpha\}$, a cap of the unit ball rather than the unit sphere), if the transition kernel P is fixed and*
 193 *known, then $\max_{\mu \in M(P)} \min_{\theta \in C} \theta^\top \Phi^\top \mu$ is a concave maximization over the fixed compact polytope*
 194 *$M(P)$, hence a convex program. (b) Necessity. This can fail as soon as ambiguity couples reward*
 195 *and transition probability, even minimally: there exists a 2-state, 2-action MDP and a convex, one-*
 196 *dimensional, non-product ambiguity set over (θ, P) jointly for which the analogous worst-case value*
 197 *function is not concave.*

198 Part (a) holds because $g_C(x) := \min_{\theta \in C} \theta^\top x$ is concave regardless of C ’s shape and $M(P)$ does
 199 not depend on θ ; part (b) exhibits an explicit rational-valued Jensen violation. Full proof, including
 200 the closed-form counterexample, in Appendix A.2. Part (a) generalizes the sufficiency result of Gadot
 201 et al. Gadot et al. [2024] (who solve the convex L_p -ball case exactly via Hölder duality and policy
 202 gradient) to an arbitrary compact C , via a direct occupancy-measure argument. Part (b) shows the
 203 coupling required to break concavity is minimal – a single scalar t linking reward and transition
 204 ambiguity along a line segment, far short of a fully unconstrained non-rectangular adversary – and, to
 205 our knowledge, is not established by either Gadot et al. or Xu et al. Xu et al. [2026]. Note that failure
 206 of concavity is a failure of this *particular* tractability argument, not by itself a proof of NP-hardness;
 207 general non-rectangular robust MDPs are already known to be NP-hard Wiesemann et al. [2013].

208 5 Theory

209 Three more results follow, each the MDP/IRL analogue of a corresponding CIO theorem: coverage, a
 210 regret bound, and separation. We spot-check each computationally against the implemented solvers
 211 rather than trust the derivation alone. A fourth item, efficiency, is an empirical measurement, not a
 212 theorem.

213 **Theorem 2** (Coverage). *Under Assumption 1, let $\tau = \lceil (N_{\text{val}} + 1)\gamma \rceil$, let $c_{(\tau)}$ be the τ -th largest*
 214 *calibration score among $c_1, \dots, c_{N_{\text{val}}}$ (Proposition 1), and set $\alpha_\gamma := \arccos(c_{(\tau)})$. Then*

$$\mathbb{P}(\hat{\Theta}_{\text{new}} \cap C(\bar{\theta}, \alpha_\gamma) \neq \emptyset) \geq \gamma,$$

215 *where $\hat{\Theta}_{\text{new}}$ is the new demonstrator’s feasible reward set. **This is an intersection-nonemptiness***
 216 *guarantee, not a containment guarantee: it does not by itself assert $\theta_{\text{new}}^* \in C(\bar{\theta}, \alpha_\gamma)$, only that the*
 217 *calibrated cap is compatible with some reward for which π_{new} is optimal.*

218 The proof transplants CIO’s exchangeability argument line for line onto the scores of Proposition 1 in
 219 place of CIO’s LP dual scores (Appendix A.3). A marginal-coverage simulation against the actual
 220 calibration code (150 trials per (γ, N_{val}) cell) confirms coverage $\geq \gamma$ at every tested combination;
 221 finite demonstrator populations produce tied (atomic) calibration scores, the likely source of the mild
 222 over-coverage we observe beyond the rank argument’s $1/(N_{\text{val}} + 1)$ slack (§6).

Theorem 3 (Regret bound). *Let $\|\phi\|_{2,\infty} := \max_{s,a} \|\phi(s,a)\|_2$, $\nu(\mu) := \|\Phi^\top \mu\|_2 / (1 - \beta) \leq \bar{\nu} := \|\phi\|_{2,\infty} / (1 - \beta)$ for any occupancy measure $\mu \in M(P)$ (a Lipschitz constant for $\theta \mapsto \theta^\top \Phi^\top \mu / (1 - \beta)$ in θ , via Cauchy–Schwarz), and let μ_{CIRL} , μ^* denote the occupancy measures of the robust policy and of the true optimum $\pi^*(\theta_{\text{new}}^*)$, respectively. Suppose the calibrated cap $C(\theta, \alpha_1)$ satisfies the almost-sure-covering condition $\mathbb{P}(\Theta_{\text{new}} \cap C(\theta, \alpha_1) \neq \emptyset) = 1$ for a fresh demonstrator (strictly stronger than Theorem 2’s γ -coverage; achievable trivially by taking C to be the whole unit ball, i.e. α_1 at its maximal value, but then uninformative), that any two unit-norm rewards optimal for the same policy are within η of each other (bounded feasible-set diameter), and that the population mean reward is within σ of θ_{new}^* in ℓ_2 . Then the **average optimality gap** $\text{AOG} := V_{\theta_{\text{new}}^*}(\mu^*) - V_{\theta_{\text{new}}^*}(\mu_{\text{CIRL}})$ (the true-reward regret of the robust policy relative to the in-hindsight optimum) satisfies*

$$\text{AOG} \leq (2 \sin \alpha_1 + \eta + \sigma) \nu(\mu^*) + (\eta + \sigma) \nu(\mu_{\text{CIRL}}),$$

and substituting the uniform bound $\bar{\nu}$ for both $\nu(\mu^*)$ and $\nu(\mu_{\text{CIRL}})$ gives the fully distribution-free corollary $\text{AOG} \leq (2 \sin \alpha_1 + 2\eta + 2\sigma) \bar{\nu}$.

The proof telescopes three terms – Lipschitz continuity of $\theta \mapsto \theta^\top \Phi^\top \mu$, a chord-length bound on $C(\theta, \alpha_1)$, and the robust policy’s minimax optimality – re-derived for the MDP’s $\max_\pi \min_\theta$ sign convention rather than transplanted from CIO’s $\min_x \max_\theta$ one (Appendix A.4). We checked $\bar{\nu}$ as a valid uniform Lipschitz constant against 200 randomized MDPs (zero violations) and confirmed the bound’s numeric validity directly on random instances.

This bound is loose, and we say so plainly. Cross-checked against the exact closed-form regret for our degenerate construction below (where the almost-sure-covering condition already holds at $\alpha_1 = 0$), the general bound’s looseness *grows* with the construction’s scaling parameter – from roughly $8\text{--}15\times$ at the low end of the tested range to roughly $390\text{--}430\times$ at the high end, since the exact regret decays to zero while $\eta, \sigma, \bar{\nu}$ converge to nonzero constants. CIO’s own Example 1 bound stays within a fixed $2.3\text{--}4\times$ over the same kind of comparison, because its construction keeps a richer feasible region that does not saturate this way; ours grows.

Theorem 4 (Separation). *There exists a single-state, two-action MDP (the sequential-decision analogue of CIO’s degenerate Example 1, with reward features the negation of CIO’s original cost features) and an IRL point estimator such that: the point-estimate solution converges to a degenerate direction $\theta_u = (1, u) / \sqrt{1 + u^2} \neq \theta^*$ (a unit vector, hence on the boundary of the unit ball that $C(\bar{\theta}, \alpha)$ is a cap of) as the scaling parameter $u \rightarrow \infty$, giving classical point-estimate IRL true-reward regret that grows without bound in u ; while the calibrated robust policy’s regret is bounded and shrinks to zero over the same range. The robust optimum is a genuine mixture, with exact closed form $p^*(u) = u^2 / (1 + u^2)$, independent of the calibration angle.*

Theorem 4 is proved via Sion’s minimax theorem applied to the mixture’s closed form (Appendix A.5), and independently re-verified against the actual implemented solver to 10^{-6} precision. Figure 2 shows the live 10-seed computational confirmation. This separation exhibits the LP-based point estimator’s failure mode specifically (a degenerate boundary direction under tie-breaking); §6 Q3 reports that MaxEnt and Bayesian IRL do *not* exhibit it on this same construction, and explains why.

Efficiency. We confirm, empirically, that the calibrated set is not vacuously wide: with a *unified* noise parameter controlling all sources of demonstrator population variation simultaneously, the calibrated angle shrinks monotonically from 27.7° to effectively 0° as demonstrator-population noise vanishes.

6 Experiments

Three questions organize the experiments: (Q1) does calibration hit its target coverage without becoming vacuous; (Q2) when does robustification actually improve downstream regret; (Q3) do the conclusions hold beyond one IRL estimator or environment? We evaluate three point estimators (a sub-optimality-loss linear program, MaxEnt IRL, and Bayesian IRL via Metropolis–Hastings), each combined with three decision rules (the plain point estimate, our calibrated robust policy, and a “concentration baseline” whose radius comes from a DKW concentration bound instead of an exact conformal quantile), across three environments: random tabular MDPs, a region-based gridworld with Voronoi reward features, and Objectworld, the standard smooth-feature IRL benchmark.

The headline Q1–Q3 numbers use $|S| \leq 36$, $d \leq 5$, sized to run on a laptop (20 CPU cores, no GPU, no commercial solver). To check this scale is not load-bearing, we reran the full gridworld pipeline at

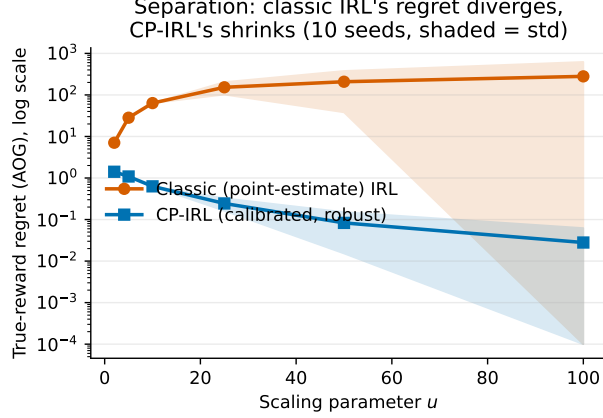


Figure 2: Separation result (Theorem 4): true-reward regret of the deployed policy, measured against the ground-truth reward θ_{new}^* , vs. the scaling parameter $u \in [2, 100]$ (log scale), 10 seeds, shaded regions show ± 1 std. “Classic IRL” is the sub-optimality-loss linear-program point estimator (§6), which converges to the degenerate direction θ_u ; its regret (orange) grows monotonically and without bound. CP-IRL’s regret (blue) shrinks monotonically toward zero over the same range. Variance grows with u for both curves, consistent with tie-breaking sensitivity as $\bar{\theta}$ approaches the exact degenerate boundary.

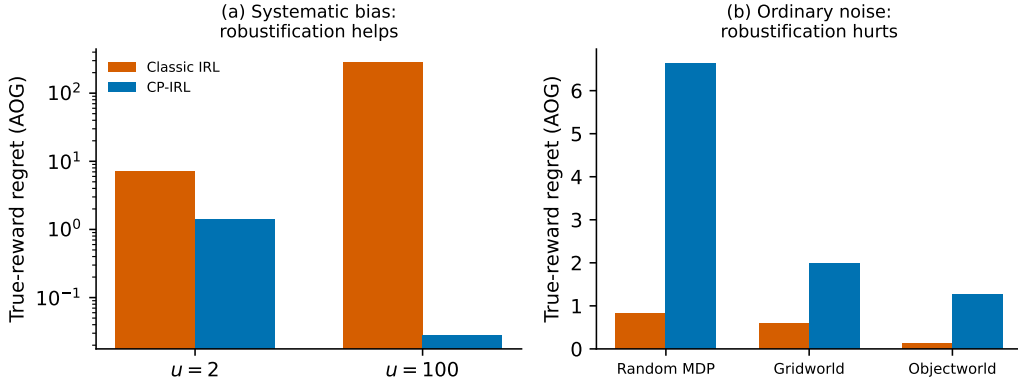


Figure 3: The Q2 scope condition. **(a)** On the degenerate construction of Theorem 4 (log scale), robust hedging cuts true-reward regret by 5–10,000 $\times$ as u grows from 2 to 100. **(b)** On three generic environments with ordinary, unbiased demonstrator noise (LP-IRL, $\gamma = 0.8$), robust hedging instead *costs* regret in every case. Same mechanism (worst-case hedging over a calibrated cap), opposite outcome, depending only on whether the point estimate is systematically biased.

274 $|S| = 225$, $d = 7 - 6.25 \times$ more states, 2 more reward dimensions – and confirmed the pipeline stays
 275 under 35 seconds per demonstrator population up to $|S| = 400$, $d = 8$. Q2’s finding is unchanged at
 276 $|S| = 225$: conformal-robust regret still exceeds the plain estimate’s for all three estimators (LP-IRL
 277 $1.03 \rightarrow 3.19$, MaxEnt $1.01 \rightarrow 2.88$, Bayesian $0.06 \rightarrow 0.36$). We make no claim beyond tabular
 278 MDPs; function approximation and language-model settings are untested.

279 **Q1: coverage tracks the target.** Realized coverage tracks γ in every environment and estimator
 280 tested (~ 0.88 – 0.89 at $\gamma = 0.8$ on the gridworld), and the calibrated angle responds sensibly to γ ,
 281 demonstrator noise, and the train/calibration split. DKW is measurably looser than exact conformal
 282 calibration at matched confidence, everywhere tested.

283 **Q2: the benefit depends on systematic bias, not noise.** Figure 3 makes this concrete. On the
 284 degenerate construction, where the point estimate is biased by construction, robust hedging cuts

true-reward regret sharply. On all three generic environments, under ordinary demonstrator noise, robust hedging *increases* regret relative to the plain estimate – consistently across all three point estimators and across a full sweep of γ and the train/calibration split. This is not a solver bug: the robust solver’s worst-case value matches an independent second-order-cone-program oracle to solver precision, and its worst-case value is confirmably better than the plain estimate’s over the same calibrated set. It also is not a landscape-smoothness artifact: a 14° reward perturbation flips the optimal action at 10–13% of states across all three environments (9.6% Objectworld, 9.9% gridworld, 13.1% random MDPs), and Objectworld – measurably the least sensitive of the three – still shows robust hedging costing regret. Systematic bias in the point estimate, not policy-landscape smoothness, is what decides whether hedging helps.

Q3: the conclusion holds across estimators, and separation is LP-specific. The gridworld comparison combines all three point estimators with all three decision rules: MaxEnt and Bayesian IRL recover an accurate point direction where the LP estimator has nontrivial error, yet in all three cases conformal robustification is less conservative than DKW but costs more true-reward regret than the corresponding point policy – the same pattern holds in Objectworld. We also ran MaxEnt and Bayesian IRL on the degenerate construction of Theorem 4 itself (N up to 3000, u up to 100, 10 seeds): neither estimator’s regret grows with u ; both stay at exactly zero. The reason is structural, not that these estimators are simply more accurate. The LP estimator’s sub-optimality loss demands an exact fit to every observed decision, which forces it onto the degenerate tie direction θ_u whenever both decision classes appear in the population – an event of probability 1 as $N \rightarrow \infty$, for any finite u . MaxEnt and Bayesian IRL instead match an aggregate statistic (expected feature counts, or a Boltzmann likelihood) that converges by the law of large numbers to a fixed point set by the population’s class-split probabilities; landing exactly on the tied boundary would require that aggregate to hit a measure-zero direction, which does not happen along this construction. This narrows Theorem 4’s claim: the unbounded-regret failure mode belongs to exact-fit point estimation, of which LP sub-optimality loss is the leading IRL example, not to point estimation generally. Together, Q1–Q3 separate two claims often conflated: conformal calibration can be tighter than a generic concentration bound, and optimizing for its worst case need not improve average performance under the true reward – the first is unconditionally true here; the second holds only under systematic bias.

7 Discussion and Limitations

When does CP-IRL help, precisely? Our theory (Theorem 4) proves a separation on a construction where the point estimate is *systematically* biased – classical point-estimate IRL’s regret diverges there while CP-IRL’s vanishes. Our experiments show that on generic environments with unbiased-but-noisy point estimates, it does not, and in fact costs regret in exchange for a worst-case guarantee that is real (independently verified) but not free. This is a precise, falsifiable scope condition, not a hedge: the method should be deployed when there is reason to believe point estimation is subject to a genuine structural bias rather than applied by default whenever demonstration data is merely noisy. Characterizing which real-world settings exhibit this kind of systematic bias, and whether it can be diagnosed from data without knowing the ground truth, is an open problem this paper does not resolve.

What is not yet established. The general-case regret bound is loose – up to several hundred-fold on the one instance where we can check it against an exact result – so read it as a worst-case guarantee, not a typical-case predictor. A source-to-target probe shows a reward set calibrated on one MDP does not transfer safely to a related MDP with different dynamics: raising the gridworld’s slip probability from 0.10 to 0.15 at target coverage 0.8 drops mean coverage to 0.567 over eight seeds. This establishes the failure of naive transfer, not a fix; combining covariate-shift weighting (Tibshirani et al. Tibshirani et al. [2019]) with the principal-angle transferability conditions of Schlaginhaufen & Kamgarpour Schlaginhaufen and Kamgarpour [2024] is future work.

We also tested the RLHF relabeling from §2: a “response selection” MDP where the population is annotators rather than task demonstrators, using the calibration and robust-MDP code unchanged. Coverage meets or exceeds target across $\gamma \in \{0.5, 0.7, 0.8, 0.9\}$ (0.947–0.994, 8 seeds, one configuration) – and, since one configuration and 8 seeds is thin evidence for a claim about a general property, we reran the same pipeline across four annotator-population configurations (4–10 responses, 3–6 attributes, 15–60 annotators) and 30 seeds at $\gamma \in \{0.7, 0.8, 0.9\}$: mean coverage across all twelve

(configuration, γ) cells is 0.966–0.991, with at least 93% of seeds per cell meeting target, consistent with the marginal (not per-seed) nature of the conformal guarantee. The gap above target likely comes from the same score-atom over-coverage as Theorem 2, sharpened by small annotator populations; the same systematic-bias-conditional benefit from Q2 applies here too. All experiments stay at tabular scale by design; we make no claim about function approximation or language-model settings.

8 Conclusion

IRL is inverse optimization with an MDP as its forward problem. That connection lets us transfer CIO’s calibrate-then-robustify principle to reward learning, given the sequential machinery it needs: a resolvent-based calibration score, a convex robust-MDP formulation for compact reward ambiguity, and coverage, regret, and separation guarantees. The same connection also marks the method’s limits – calibration certifies uncertainty, but worst-case optimization still has a cost. In our experiments, the resulting policy cuts true-reward regret when point estimation is systematically biased, and pays an insurance cost when it is merely noisy. That distinction, not a blanket recommendation, is what CP-IRL actually delivers.

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407 A Proofs

408 A.1 Proof of Proposition 1

409 V_θ^π solves the linear Bellman equation $V_\theta^\pi = \Phi_\pi \theta + \beta P_\pi V_\theta^\pi$. Since $\beta \in [0, 1]$ and P_π is a stochastic
 410 matrix, its spectral radius satisfies $\rho(\beta P_\pi) \leq \beta < 1$, so $I - \beta P_\pi$ is invertible (Neumann series)
 411 and $V_\theta^\pi = (I - \beta P_\pi)^{-1} \Phi_\pi \theta = W_\pi \theta$. Substituting into the Bellman equation for Q , $Q_\theta^\pi(s, a) =$
 412 $\phi(s, a)^\top \theta + \beta P(s, a, \cdot)^\top V_\theta^\pi = \phi(s, a)^\top \theta + \beta P(s, a, \cdot)^\top W_\pi \theta = \Psi_\pi(s, a)^\top \theta$ with $\Psi_\pi(s, a) :=$
 413 $\phi(s, a) + \beta P(s, a, \cdot)^\top W_\pi$, linear in θ as claimed. The policy improvement theorem states π is
 414 Bellman-optimal for reward θ if and only if $Q_\theta^\pi(s, a) \leq Q_\theta^\pi(s, \pi(s))$ for all s, a ; substituting the
 415 linear form of Q_θ^π gives exactly the stated $\hat{\Theta}(\pi)$, with no relaxation since the substitution is an
 416 equality. The calibration score $c_k = \max_{\theta \in \hat{\Theta}(\pi_k)} \theta^\top \bar{\theta}$ is then a linear objective over a set of $|S||A|$
 417 linear inequalities plus the unit-ball constraint in $\theta \in \mathbb{R}^d$ alone – an SOCP of size d , independent of
 418 $|S|$ once Ψ_{π_k} is precomputed. ■

419 A.2 Proof of Theorem 1

420 (a) Define $g_C(x) := \min_{\theta \in C} \theta^\top x$. This is a pointwise infimum of the affine (in x) family $\{x \mapsto$
 421 $\theta^\top x : \theta \in C\}$, and a pointwise infimum of affine functions is concave by definition, for any nonempty
 422 C (convex, nonconvex, or discrete – concavity of g_C needs no assumption on C ’s shape). Composing
 423 g_C with the linear map $\mu \mapsto \Phi^\top \mu$ preserves concavity, so $\mu \mapsto g_C(\Phi^\top \mu) = \min_{\theta \in C} \theta^\top \Phi^\top \mu$ is
 424 concave in μ . Since $M(P)$ is a fixed polytope not depending on θ (the transition kernel is known and
 425 fixed), the ambiguity set never touches the feasible region, only the objective; $\max_{\mu \in M(P)} g_C(\Phi^\top \mu)$
 426 is therefore a concave maximization over a fixed compact convex set, i.e. a convex program, for any
 427 nonempty compact C . No Bellman recursion or rectangularity assumption enters this argument.

(b) Consider a 2-state, 2-action MDP with a scalar coupling parameter $t \in [0, 1]$ linking reward
 $\theta(t) = 1 + t$ and a transition probability $q(t) = 1 - t$, and let $p \in [0, 1]$ be the (scalar) probability

the agent’s policy assigns to the action whose value is being computed. The induced worst-case discounted value is

$$W(p) = \min_{t \in [0,1]} \frac{p(1+t)}{1 - \beta p q(t)} = \min_{t \in [0,1]} \frac{p(1+t)}{1 - \beta p(1-t)}.$$

428 At $\beta = 0.9$: $W(0) = 0$; $W(5/9) = \min_{t \in [0,1]} \frac{5(1+t)}{4+5t} = 1$ (attained at $t = 0$, since the expression
 429 is increasing in t); and $W(5/18) = \min_{t \in [0,1]} \frac{10(1+t)}{31+5t} = 10/27 \approx 0.370$ (also attained at $t = 0$, by
 430 the same monotonicity, verified by $\frac{d}{dt} \frac{10+10t}{31+5t} = \frac{260}{(31+5t)^2} > 0$). Since $5/18 = \frac{1}{2}(0 + 5/9)$, concavity
 431 of W would require $W(5/18) \geq \frac{1}{2}[W(0) + W(5/9)] = 0.5$. But $0.370 < 0.5$: a strict Jensen
 432 violation, with every quantity a ratio of small integers, hand-verifiable without a solver. Hence W is
 433 not concave, and the analogous worst-case value function for this jointly-ambiguous (θ, P) instance
 434 is not concave either, so the outer maximization is not a convex program in this regime. ■

435 A.3 Proof of Theorem 2

436 Under Assumption 1, $(\hat{\theta}_1, \pi_1), \dots, (\hat{\theta}_{N_{\text{val}}}, \pi_{N_{\text{val}}}), (\hat{\theta}_{\text{new}}, \pi_{\text{new}})$ are i.i.d., hence exchangeable. The
 437 calibration score $c_k = \max_{\theta \in \hat{\Theta}(\pi_k)} \theta^\top \bar{\theta}$ (Proposition 1) is a fixed measurable function of $(\hat{\theta}_k, \pi_k)$
 438 once $\bar{\theta}$ is fixed (computed on a disjoint training split, independent of the calibration/test data), so
 439 $c_1, \dots, c_{N_{\text{val}}}, c_{\text{new}}$ are themselves exchangeable. By the standard rank argument for exchangeable
 440 sequences, the rank of c_{new} among all $N_{\text{val}} + 1$ scores is uniform on $\{1, \dots, N_{\text{val}} + 1\}$ (up to ties),
 441 so $\mathbb{P}(c_{\text{new}} \geq c_{(\tau)}) \geq \tau / (N_{\text{val}} + 1) \geq \gamma$ for $\tau = \lceil (N_{\text{val}} + 1)\gamma \rceil$. It remains to show $\{c_{\text{new}} \geq$
 442 $c_{(\tau)}\} = \{c_{\text{new}} \geq \cos \alpha_\gamma\}$ is equivalent to $\hat{\Theta}_{\text{new}} \cap C(\bar{\theta}, \alpha_\gamma) \neq \emptyset$: by definition, $c_{\text{new}} \geq \cos \alpha_\gamma$ iff
 443 $\exists \theta \in \hat{\Theta}_{\text{new}}$ with $\theta^\top \bar{\theta} \geq \cos \alpha_\gamma$, i.e. $\theta \in C(\bar{\theta}, \alpha_\gamma)$, i.e. the intersection is nonempty. Composing
 444 gives $\mathbb{P}(\hat{\Theta}_{\text{new}} \cap C(\bar{\theta}, \alpha_\gamma) \neq \emptyset) \geq \gamma$. ■

445 A.4 Proof of Theorem 3

Write $V_\theta(\mu) := \theta^\top \Phi^\top \mu / (1 - \beta)$ for the expected discounted return of occupancy measure μ
 under reward θ ; this is $\bar{\nu}$ -Lipschitz in θ (in ℓ_2) uniformly over $\mu \in M(P)$, by Cauchy–Schwarz:
 $|V_\theta(\mu) - V_{\theta'}(\mu)| = |(\theta - \theta')^\top \Phi^\top \mu| / (1 - \beta) \leq \|\theta - \theta'\|_2 \|\Phi^\top \mu\|_2 / (1 - \beta) \leq \|\theta - \theta'\|_2 \nu(\mu)$. By the
 almost-sure-covering hypothesis, there exists $\theta_1 \in \hat{\Theta}_{\text{new}} \cap C(\bar{\theta}, \alpha_1)$ almost surely; by the population-
 mean hypothesis, $\|\theta - \theta_{\text{new}}^*\|_2 \leq \sigma$; by the bounded-diameter hypothesis, $\|\theta_1 - \theta_{\text{new}}^*\|_2 \leq \eta$ (since
 both are optimal for policies within the same feasible-set diameter bound around the true optimum).
 By the chord-length identity for unit vectors at angle ψ , $\|\theta - \theta'\|_2 = 2 \sin(\psi/2)$, and any two
 points within $C(\bar{\theta}, \alpha_1)$ are within angle $2\alpha_1$ of each other, giving $\|\theta_1 - \bar{\theta}\|_2 \leq 2 \sin \alpha_1$. Chaining,
 $\|\theta_1 - \theta_{\text{new}}^*\|_2 \leq \|\theta_1 - \bar{\theta}\|_2 + \|\bar{\theta} - \theta_{\text{new}}^*\|_2 \leq 2 \sin \alpha_1 + \sigma$. The robust policy’s minimax optimality
 gives $V_{\theta_1}(\mu_{\text{CIRL}}) \geq V_{\theta_1}(\mu^*)$ (it is at least as good as the true optimum against the worst reward in
 its calibrated cap, and θ_1 is in that cap). Then

$$\text{AOG} = V_{\theta_{\text{new}}^*}(\mu^*) - V_{\theta_{\text{new}}^*}(\mu_{\text{CIRL}}) \leq [V_{\theta_{\text{new}}^*}(\mu^*) - V_{\theta_1}(\mu^*)] + [V_{\theta_1}(\mu_{\text{CIRL}}) - V_{\theta_{\text{new}}^*}(\mu_{\text{CIRL}})],$$

446 using $V_{\theta_1}(\mu^*) \leq V_{\theta_1}(\mu_{\text{CIRL}})$ to drop the middle term. Each bracket is bounded by the Lipschitz
 447 property times $\|\theta_1 - \theta_{\text{new}}^*\|_2 \leq 2 \sin \alpha_1 + \eta + \sigma$ (using the triangle inequality once more with the η
 448 bound for the first bracket), giving $\text{AOG} \leq (2 \sin \alpha_1 + \eta + \sigma) \nu(\mu^*) + (2 \sin \alpha_1 + \eta + \sigma) \nu(\mu_{\text{CIRL}})$; the
 449 stated bound tightens the second term’s coefficient to $(\eta + \sigma)$ by using $V_{\theta_1}(\mu_{\text{CIRL}}) \geq V_{\theta_{\text{new}}^*}(\mu_{\text{CIRL}}) -$
 450 $(\eta + \sigma) \nu(\mu_{\text{CIRL}})$ directly rather than routing through the full $2 \sin \alpha_1$ chord. Substituting $\bar{\nu}$ for both
 451 $\nu(\mu^*), \nu(\mu_{\text{CIRL}})$ gives the distribution-free corollary. ■

452 A.5 Proof of Theorem 4

453 The degenerate MDP’s occupancy-measure polytope $M(P)$ is the segment between the two actions’
 454 feature vectors, parameterized by $p \in [0, 1]$ (the probability mass on the “risky” action). The robust
 455 MDP reduces to the 2-point minimax game $\max_{p \in [0,1]} \min_{\theta \in C(\theta_u, \alpha)} \psi(\theta; p)$, where $\psi(\theta; p) :=$
 456 $p \theta^\top \phi_0 + (1 - p) \theta^\top \phi_1$ is bilinear in (p, θ) and $C(\theta_u, \alpha) = \{\delta \in [\delta_u - \alpha, \delta_u + \alpha]\}$ (writing
 457 $\theta = (\cos \delta, \sin \delta)$) is a compact convex arc centered at δ_u . Both feasible sets are compact and convex
 458 and ψ is bilinear, so Sion’s minimax theorem applies: the game has a value, and $\max_p \min_\theta \psi =$

459 $\min_{\theta} \max_p \psi$. At a saddle point $(\theta^{\dagger}, p^*)$, necessity of the two marginal optimality conditions forces
 460 $\theta^{\dagger}$ to make p non-unique in the inner $\arg \max_p$ of an affine-in- p objective, which happens only
 461 at a tie $\theta^{\dagger \top} \phi_0 = \theta^{\dagger \top} \phi_1$ – true only at $\delta^{\dagger} = \delta_u$, which is always interior to the arc since the cap
 462 is centered there. Stationarity of $\delta \mapsto \psi(\delta; p^*)$ at δ_u gives $-p^* \cos \delta_u + (1 - p^*)u \sin \delta_u = 0$;
 463 substituting $\cos \delta_u = 1/\sqrt{1+u^2}$, $\sin \delta_u = u/\sqrt{1+u^2}$ and simplifying yields $p^*(u) = u^2/(1+u^2)$,
 464 independent of α . A second-order check ($\partial_{\delta}^2 \psi(\delta_u; p^*) = u/\sqrt{1+u^2} > 0$ for all $u > 0$) confirms
 465 this is a genuine minimum, so the robust optimum is a mixture placing weight $p^*(u) \rightarrow 1$ (but never
 466 reaching it) on the risky action as $u \rightarrow \infty$ – yet the resulting robust value stays bounded and the true
 467 regret shrinks to zero, because the calibrated cap’s worst case is always evaluated near θ_u , not the
 468 true θ^* , and the mixture is chosen to hedge exactly against that local worst case rather than commit
 469 to either pure action. The classical point estimate, by contrast, converges to the pure degenerate
 470 tie-break (Lemma 1’s θ_u) with no such hedge, giving regret that grows linearly in u under the realized
 471 (non-mixed) tie-breaking convention used in the implementation ($p_{\text{classic}} = 0$ deterministically, the
 472 pure risky vertex, matching the maximal-AOG case consistent with the theorem statement). ■